THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which among the following is the best use of the sacrificial anode principle? 1

(a) zinc blocks on ship hull **(b)** painting a gate
(c) oiling a bicycle chain **(d)** making stainless steel

2. Assertion (A): Graphite is a non-metal and a conductor of electricity. Reason (R): Graphite has free electrons in its layers. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. An element has 12 protons. It forms an ion and a compound with oxygen as follows: 1

(a) charge +2, formula MgO_2 **(b)** charge +1, formula MgO
(c) charge +2, formula MgO **(d)** charge -2, formula Mg_2O

4. A metal is found as sulphide ore. The correct sequence of extraction is: 1

(a) electrolysis -> reduction **(b)** calcination -> electrolysis
(c) roasting -> reduction -> refining **(d)** reduction -> roasting

5. Assertion (A): Calcium floats on water during reaction. Reason (R): Calcium is lighter than water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Metal M with electronic configuration 2, 8, 3 forms compound with sulphur (S^{2-}): 1

(a) M_2S **(b)** MS **(c)** M_2S_3 **(d)** M_3S_2

7. Zinc blende is first roasted (and not calcined) because it is: 1

(a) a sulphide **(b)** a chloride **(c)** a carbonate **(d)** an oxide

8. Assertion (A): Pure gold is not used for making jewellery. Reason (R): Pure gold is very soft. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Metals are electropositive elements. 1
Reason (R): Metals tend to lose electrons to form positive ions.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Sodium chloride conducts electricity in the molten state but not in the solid state. 1
Reason (R): Ionic compounds have high melting points.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A metal M burns in air with a dazzling white flame giving a white powder. Identify M and write the equation. 2

12. Write the balanced equation for the thermit reaction and state one use of it. 2

13. What are the products of electrolysis of molten NaCl at the anode and cathode? 2

14. State whether the following will react and why: 2
(i) Cu and $AgNO_3$
(ii) Ag and $CuSO_4$.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the extraction of iron from haematite and of sodium from rock salt giving the method and reason. 3

16. Explain why a mixture of iron filings and sulphur is different from iron sulphide on heating. 3

17. Zinc is added to copper sulphate solution. Write the equation and state two observations. 3

OR

Explain the use of cryolite and graphite electrodes in the extraction of aluminium. 3

18. Explain how the thickness of the oxide layer on aluminium is increased and why this is useful. 3

OR

Explain how the reactivity series helps in deciding the method for extracting a metal. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Magnesium (atomic number 12) reacts with chlorine (atomic number 17) to form a compound. The compound is a white crystalline solid with a high melting point; it does not conduct electricity as a solid but conducts when molten or dissolved in water. 5
(i) How many electrons does a magnesium atom lose?
(ii) Write the formula of the compound.
(iii) Name the type of bond.
(iv) Explain why the compound conducts electricity in the molten state but not in the solid state.

20. Rusting experiment: nails kept in 5
(a) dry air
(b) boiled water with oil layer
(c) moist air. Predict rust, the reason, and give four methods of prevention.

OR

Why are metals like Pb and Zn found as sulphides/carbonates? Explain the conversion of a carbonate and a sulphide ore into the metal, giving equations. 5

END OF QUESTION PAPER — Total: 40 Marks